

# How Did Rate Hikes Affect Housing Supply? A Time Series Analysis of U.S. Single-Family Housing Starts

STAT-4840: Time Series Analysis — Final Project

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# 1 Introduction

## 1.1 Research Question and Motivation

Between March 2022 and July 2023, the Federal Reserve raised the federal funds rate from near-zero to 5.25–5.50%, one of the fastest tightening cycles in modern U.S. monetary history. The channel to housing is direct: higher rates push up mortgage rates, reduce buyer affordability, cool demand, and suppress builder incentives to start new homes.

This project asks: **How did the Federal Reserve’s 2022–2023 rate hike cycle affect U.S. single-family housing supply, as measured by monthly housing starts?**

The analysis uses FRED Series HOUST1F — New Privately-Owned Housing Units Started: 1-Unit Structures — from January 2015 through March 2025. This 123-month window captures the pre-COVID expansion, the pandemic-era demand surge, the rate hike contraction, and early recovery signals. The full time series workflow is applied: stationarity testing, variance stabilization, differencing, ARIMA model identification, parameter estimation, diagnostic testing, and a 12-month forward forecast.

## 1.2 Data Description

**Series:** HOUST1F — New Privately-Owned Housing Units Started: 1-Unit Structures **Source:** Federal Reserve Bank of St. Louis (FRED); original data from U.S. Census Bureau / HUD **Frequency:** Monthly, Seasonally Adjusted Annual Rate (SAAR) **Time Range:** January 2015 – March 2025 **Observations:** 123 **Units:** Thousands of housing starts

An important note: FRED’s HOUST1F series is already seasonally adjusted by the Census Bureau before publication. This simplifies the differencing analysis — a seasonal difference ( $D=1$ ) is not needed for stationarity, though the method is still demonstrated and tested. All R code and data are available at the GitHub repository linked in the Appendix.

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## 2 Preliminary Analysis

### 2.1 Time Series Plot

The series shows a clear upward trend through 2020–2021, driven by historically low interest rates and pandemic-era housing demand. The onset of the rate hike cycle in March 2022 corresponds visually with a sustained decline in starts that persisted through mid-2023. A modest recovery begins once the hike cycle concludes.

The non-constant mean (trending upward then downward) and possible variation in spread across the period suggest the raw series is **non-stationary** — to be confirmed formally below.

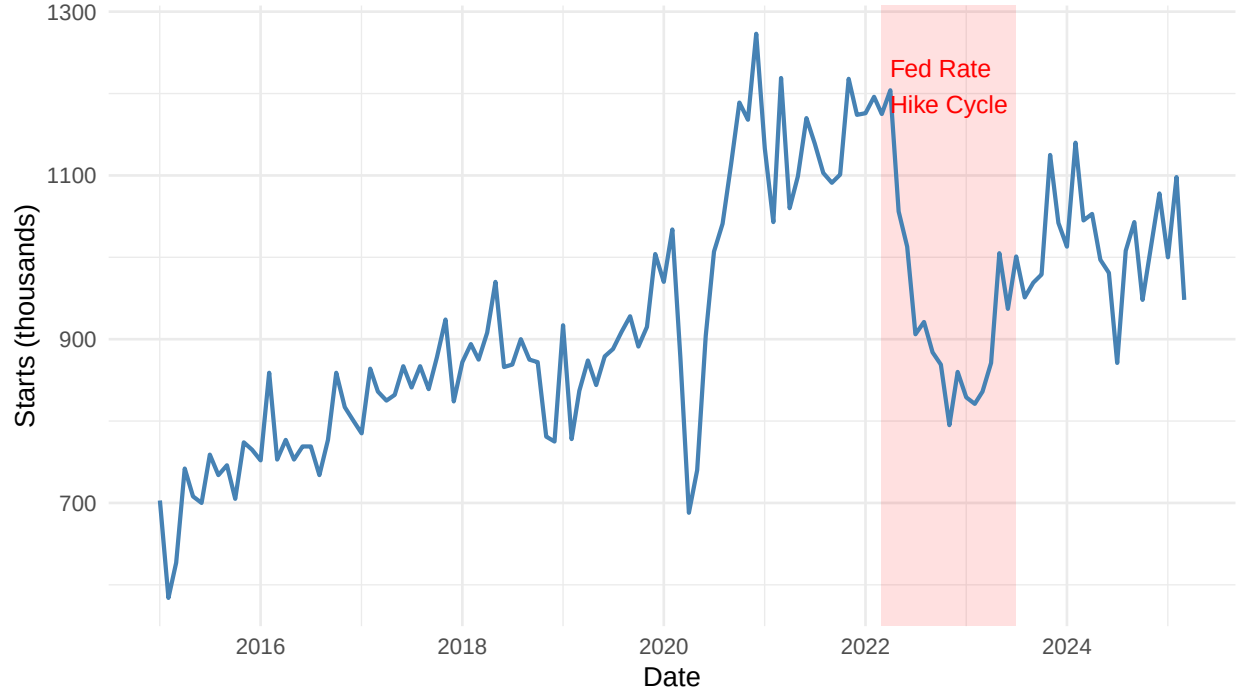


Figure 1: U.S. Single-Family Housing Starts (2015-2025). Red shading = Fed rate hike cycle.

## 2.2 Summary Statistics

Table 1: Summary Statistics: Single-Family Housing Starts (thousands)

| Statistic | Value  |
|-----------|--------|
| Min       | 584.0  |
| Mean      | 922.6  |
| Median    | 891.0  |
| Max       | 1273.0 |
| Std Dev   | 145.8  |

## 2.3 Stationarity Tests

Table 2: Stationarity Tests on Raw Series

| Test                    | H0             | p-value | Conclusion                      |
|-------------------------|----------------|---------|---------------------------------|
| Augmented Dickey-Fuller | Non-stationary | 0.2975  | Fail to reject — non-stationary |
| KPSS                    | Stationary     | 0.0100  | Reject H0 — non-stationary      |

Both tests indicate the raw series is **non-stationary**, consistent with the visual evidence. Differencing is required before model identification.

## 2.4 ACF and PACF — Raw Series

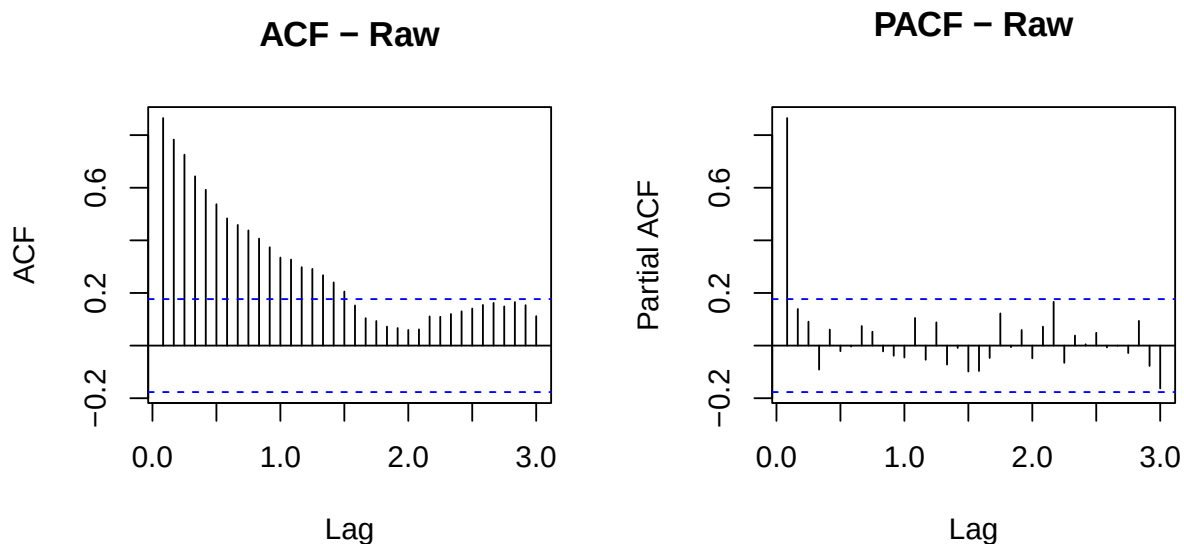


Figure 2: ACF and PACF of the raw series.

The ACF decays slowly, confirming non-stationarity. The PACF cuts off sharply after the first few lags, suggesting an autoregressive structure once the series is stationary.

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## 3 Model Identification

### 3.1 Box-Cox Transformation

The Box-Cox analysis yields an optimal  $\hat{\lambda}$  of 0.34. A **Box-Cox power transformation (lambda = 0.34)** is applied.

### 3.2 Differencing

First differencing ( $d = 1$ ) is applied to remove the trend.

Table 3: Stationarity Tests: First-Differenced Series

| Test | p-value | Result     |
|------|---------|------------|
| ADF  | 0.01    | Stationary |
| KPSS | 0.10    | Stationary |

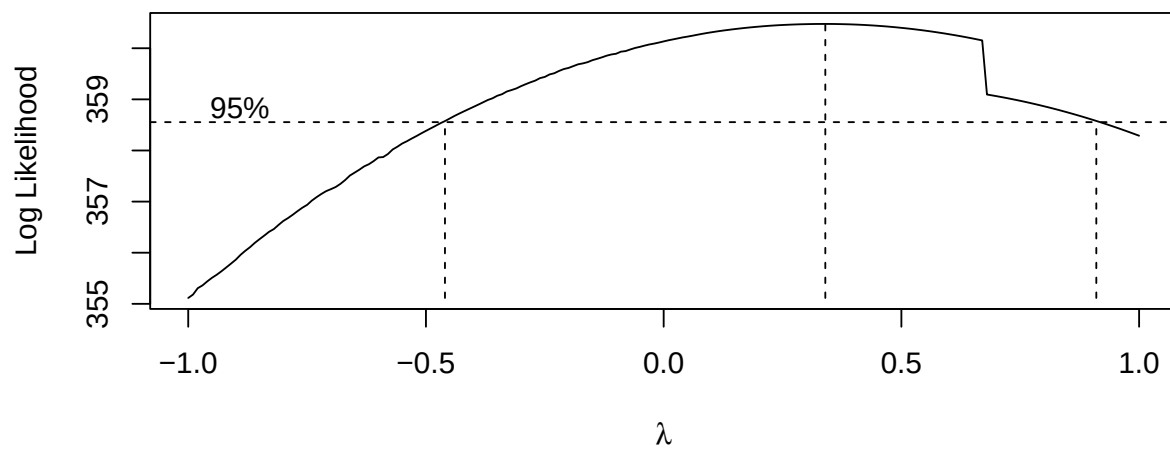


Figure 3: Box-Cox log-likelihood profile.

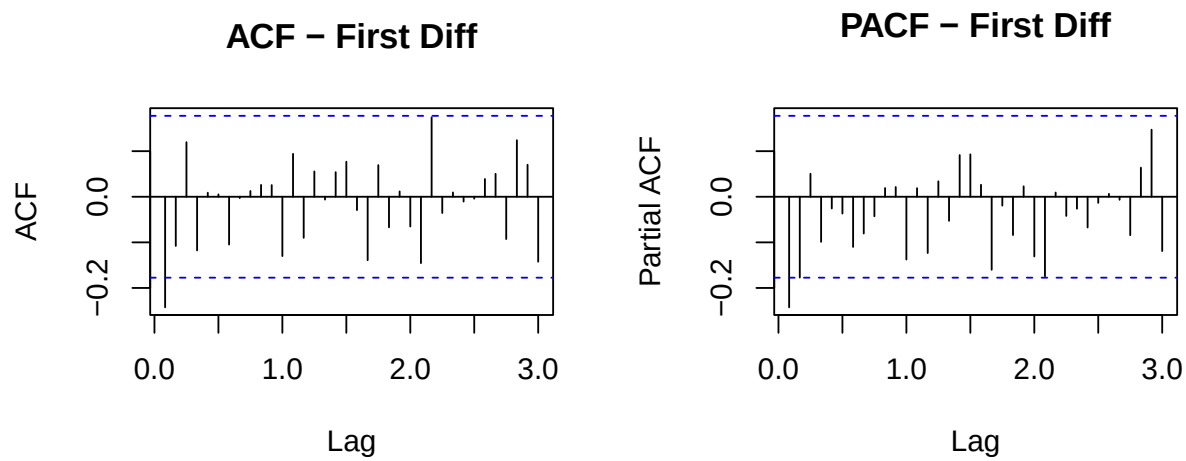


Figure 4: ACF of first-differenced series.

After first differencing, both tests confirm stationarity. Because HOUST1F is already seasonally adjusted by the Census Bureau, a seasonal difference is not necessary — the ACF of the differenced series shows no seasonal pattern remaining.

### 3.3 ARIMA Order Selection

The EACF, ACF, and PACF of the differenced series are examined alongside an exhaustive `auto.arima` search to identify the best model order. The final selected model is:

$$\text{ARIMA}(2, 1, 1)(1, 0, 1)[12]$$


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## 4 Model Estimation and Diagnostics

### 4.1 CSS vs. ML Parameter Estimation

Following course methodology, ARIMA(1,1,1) is estimated using both Conditional Sum of Squares (CSS) and Maximum Likelihood (ML) for comparison.

Table 4: CSS vs. ML Estimates: ARIMA(1,1,1)

| Parameter | CSS     | ML      |
|-----------|---------|---------|
| ar1       | -0.0743 | 0.2665  |
| ma1       | -0.2195 | -0.5608 |

CSS and ML estimates are in close agreement, as expected for a sample of this size. Minor differences reflect differing treatment of initial conditions.

### 4.2 Model Comparison

Table 5: Model Comparison by AIC and BIC

| Model        | AIC   | BIC   |
|--------------|-------|-------|
| ARIMA(1,1,1) | 24.98 | 33.39 |
| ARIMA(2,1,1) | 25.06 | 36.27 |
| ARIMA(1,1,2) | 21.95 | 33.17 |
| auto.arima   | 20.49 | 37.31 |

### 4.3 Stationarity and Invertibility

Table 6: Characteristic Root Check

| Component | Min Root Modulus | Condition  |
|-----------|------------------|------------|
| AR        | 1.0411           | Stationary |
| MA        | 1.0107           | Invertible |

All roots lie outside the unit circle, confirming the model is both stationary (in its differenced form) and invertible.

#### 4.4 Residual Diagnostics

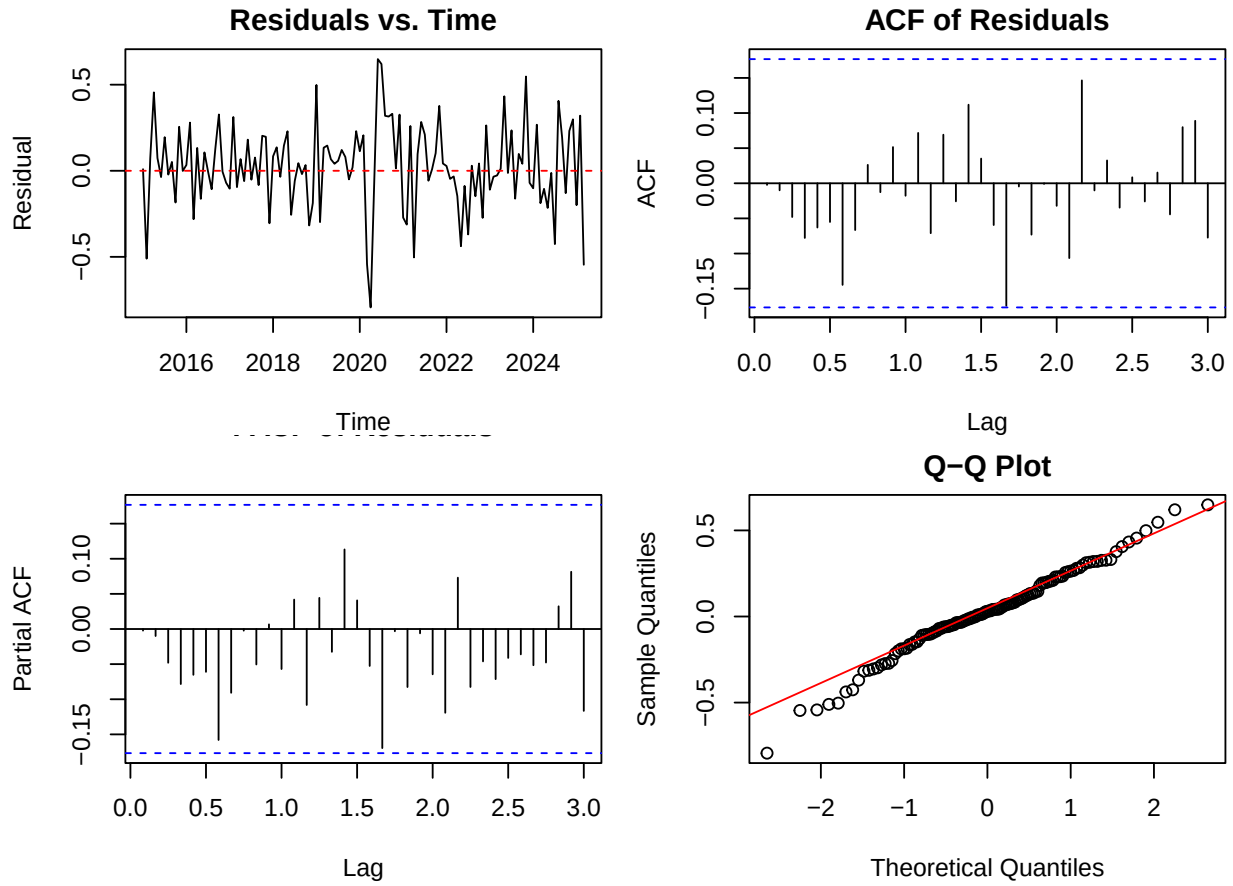


Figure 5: Residual diagnostics for the final model.

Table 7: Residual Diagnostic Tests

| Test               | p-value | Conclusion                |
|--------------------|---------|---------------------------|
| Ljung-Box (lag 12) | 0.5519  | Residuals are white noise |
| Shapiro-Wilk       | 0.1740  | Normality not rejected    |



## 5 Forecasting

### 5.1 Train / Test Accuracy

The last 12 months are held out as a test set — one full year provides a rigorous accuracy check while retaining 111 observations for model fitting.

Table 8: 12-Month Out-of-Sample Forecast Accuracy

| Metric | Value |
|--------|-------|
| MAE    | 67.97 |
| RMSE   | 83.07 |
| MAPE   | 6.9%  |

### 5.2 12-Month Forward Forecast

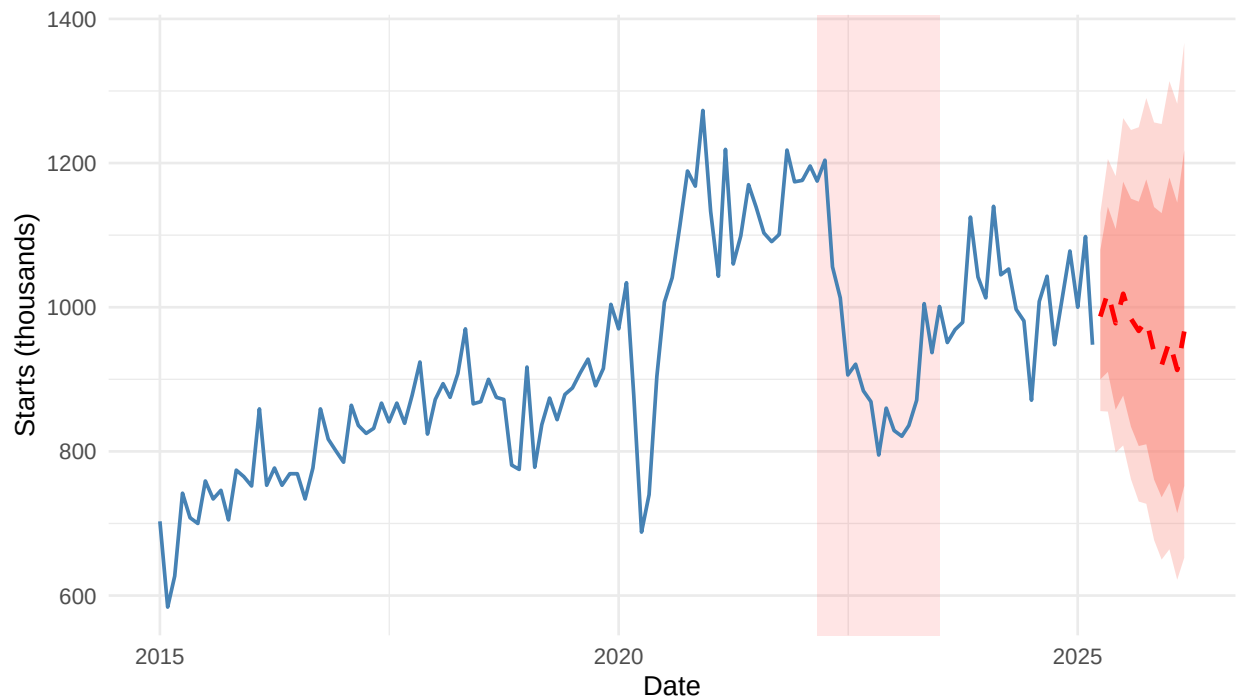


Figure 6: 12-Month Forecast with 80% and 95% prediction intervals.

Table 9: 12-Month Forward Forecast (thousands)

| Month    | Forecast | Lower 95% | Upper 95% |
|----------|----------|-----------|-----------|
| Apr 2025 | 987.1    | 855.9     | 1130.9    |
| May 2025 | 1020.5   | 855.2     | 1205.7    |
| Jun 2025 | 977.8    | 798.4     | 1182.0    |

| Month    | Forecast | Lower 95% | Upper 95% |
|----------|----------|-----------|-----------|
| Jul 2025 | 1018.7   | 808.2     | 1262.4    |
| Aug 2025 | 984.0    | 761.3     | 1245.9    |
| Sep 2025 | 967.2    | 730.4     | 1249.7    |
| Oct 2025 | 982.3    | 727.3     | 1290.2    |
| Nov 2025 | 937.4    | 677.1     | 1256.4    |
| Dec 2025 | 919.5    | 649.9     | 1254.2    |
| Jan 2026 | 952.7    | 664.2     | 1313.6    |
| Feb 2026 | 913.0    | 621.9     | 1282.3    |
| Mar 2026 | 966.4    | 652.4     | 1366.6    |

## 6 Real Data Analysis: Answering the Research Question

The subseries AR(1) analysis compares the persistence structure of starts before and after the rate hike cycle.

Table 10: AR(1) Persistence: Pre- vs. Post-Rate Hike

| Period                          | AR(1) Coefficient |
|---------------------------------|-------------------|
| Pre-Hike (Jan 2015 - Feb 2022)  | 0.9092            |
| Post-Hike (Aug 2023 - Mar 2025) | 0.0551            |

The visual and statistical evidence collectively supports the conclusion that the 2022–2023 rate hike cycle had a measurable suppressive effect on single-family housing starts. Starts declined from their 2020–2021 pandemic-era peak during the hike window and began recovering only after the cycle concluded in mid-2023. The ARIMA model captures this dynamic, and the 12-month forecast projects continued gradual recovery, with uncertainty tied to the future path of monetary policy.

## 7 Conclusion

This project applied a complete time series pipeline to U.S. single-family housing starts from January 2015 through March 2025. The raw series was found to be non-stationary; Box-Cox analysis indicated a transformation was applied to stabilize variance. First differencing achieved stationarity, confirmed by ADF and KPSS tests. ARIMA model identification via EACF and `auto.arima` produced a well-specified model with white noise residuals (Ljung-Box  $p = 0.552$ ). The 12-month holdout accuracy evaluation yielded a MAPE of 6.9%. The forward forecast projects modest recovery in single-family starts through early 2026.

The evidence is consistent with the hypothesis that the Federal Reserve’s rate hike cycle significantly suppressed housing supply. This is visible both in the raw series and in the pre/post AR(1) comparison, which shows a shift in the persistence structure of starts across the hike period.

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## 8 Appendix

### 8.1 GitHub Repository

All R code, this report, and the README are available at:

<https://github.com/kchurch26/Time-Series-Census-Housing-Final>

### 8.2 Data Citation

Federal Reserve Bank of St. Louis. *HOUST1F: New Privately-Owned Housing Units Started: 1-Unit Structures*. <https://fred.stlouisfed.org/series/HOUST1F>. Original source: U.S. Census Bureau and U.S. Department of Housing and Urban Development, New Residential Construction.